Class 12 Mathematics

Chapter 7: Integrals

Practice Set B: step-wise solutions

20 questions **54** marks **3** levels

Each row is one line of working. The red number in the right margin is what an examiner gives for that line, as in the CBSE marking scheme (½ is half a mark). Write every line that carries a mark. ✓ ½

Level 1

8 marks

1 **MCQ** NCERT Exemplar — 1 mark

$\int e^x \left(\frac{1-x}{1+x^2}\right)^2 dx$ is equal to:

(A) $\frac{e^x}{1+x^2}+C$ **(B)** $\frac{-e^x}{1+x^2}+C$

(C) $\frac{e^x}{(1+x^2)^2}+C$ **(D)** $\frac{-e^x}{(1+x^2)^2}+C$

Formula / idea: $\int e^x\,[f(x)+f'(x)]\,dx = e^x f(x)+C$

1 Expand $(1-x)^2 = (1+x^2)-2x$

$$\left(\frac{1-x}{1+x^2}\right)^2 = \frac{1}{1+x^2} - \frac{2x}{(1+x^2)^2}$$

2 Take $f(x) = \frac{1}{1+x^2}$

$$f'(x) = -\frac{2x}{(1+x^2)^2}$$

3 So the integrand is $e^x[f(x)+f'(x)]$ ✓ 1

$$\int e^x\left[\frac{1}{1+x^2} - \frac{2x}{(1+x^2)^2}\right]dx = \frac{e^x}{1+x^2}+C$$

Answer **(A)** $\frac{e^x}{1+x^2}+C$

Watch out: Expanding $(1-x)^2$ as $1-x^2$; the middle term $-2x$ is what creates $f'(x)$.

2 MCQ NCERT Exemplar 1 mark

If $\int \frac{x^3}{\sqrt{1+x^2}}\,dx = a\left(1+x^2\right)^{3/2} + b\sqrt{1+x^2} + C$, then:

(A) $a = \frac{1}{3},\ b = 1$ **(B)** $a = -\frac{1}{3},\ b = 1$

(C) $a = -\frac{1}{3},\ b = -1$ **(D)** $a = \frac{1}{3},\ b = -1$

Formula / idea: Put the expression under the root $= t$ and write the leftover x^2 in terms of t.

1 Put $1 + x^2 = t$

$2x\,dx = dt, \qquad x^2 = t - 1$

2 Write $x^3\,dx = x^2 \cdot x\,dx$ and substitute

$$\int \frac{x^3}{\sqrt{1+x^2}}\,dx = \frac{1}{2}\int \frac{t-1}{\sqrt{t}}\,dt = \frac{1}{2}\int \left(t^{1/2} - t^{-1/2}\right)dt$$

3 Integrate

$$\frac{1}{2}\left(\frac{2}{3}t^{3/2} - 2t^{1/2}\right) + C = \frac{1}{3}t^{3/2} - t^{1/2} + C$$

4 Put back $t = 1 + x^2$ and compare ✓ 1

$$\frac{1}{3}\left(1+x^2\right)^{3/2} - \sqrt{1+x^2} + C \ \Rightarrow\ a = \frac{1}{3},\ b = -1$$

Answer **(D)** $a = \frac{1}{3},\ b = -1$

Watch out: Forgetting the $\frac{1}{2}$ from $x\,dx = \frac{dt}{2}$, which doubles both a and b.

3 MCQ NCERT Misc. Ex. Q37 1 mark

The value of $\int_0^1 \sin^{-1} x\,dx$ is:

(A) $\frac{\pi}{2}$ **(B)** $\frac{\pi}{2} - 1$ **(C)** $\frac{\pi}{2} + 1$ **(D)** 1

Formula / idea: By parts with the second function 1: $\int f(x)\,dx = xf(x) - \int xf'(x)\,dx$

1 Integrate by parts, taking $\sin^{-1} x$ as first function and 1 as second

$$\int \sin^{-1} x\,dx = x\sin^{-1} x - \int \frac{x}{\sqrt{1-x^2}}\,dx$$

2 Put $1 - x^2 = t$ in the last integral

$$\int \frac{x}{\sqrt{1-x^2}}\,dx = -\sqrt{1-x^2}$$

3 So

$$\int_0^1 \sin^{-1} x \, dx = \left[x \sin^{-1} x + \sqrt{1-x^2}\right]_0^1$$

4 Put the limits ✓ 1

$$\left(\frac{\pi}{2} + 0\right) - (0 + 1) = \frac{\pi}{2} - 1$$

Answer **(B)** $\frac{\pi}{2} - 1$

Watch out: Choosing (A) by forgetting that $\sqrt{1-x^2}$ equals 1 at the lower limit.

4 **MCQ** PYQ-style 1 mark

For $x > 0, a > 0, a \neq 1$, $\int \left(e^{x \log a} + e^{a \log x} + e^{a \log a}\right) dx$ is equal to:

(A) $a^x \log a + \frac{x^{a+1}}{a+1} + a^a x + C$

(B) $\frac{a^x}{\log a} + \frac{x^{a+1}}{a+1} + a^a x + C$

(C) $\frac{a^x}{\log a} + \frac{x^{a+1}}{a+1} + \frac{a^{a+1}}{a+1} + C$

(D) $\frac{a^x}{\log a} + a x^{a-1} + a^a x + C$

Formula / idea: $e^{\log m} = m$; $\int a^x \, dx = \frac{a^x}{\log a} + C$

1 Use $e^{p \log q} = e^{\log q^p} = q^p$ on each term

$$e^{x \log a} = a^x, \qquad e^{a \log x} = x^a, \qquad e^{a \log a} = a^a$$

2 Note: a^a is a constant, not a power of x

3 Integrate term by term ✓ 1

$$\int \left(a^x + x^a + a^a\right) dx = \frac{a^x}{\log a} + \frac{x^{a+1}}{a+1} + a^a x + C$$

Answer **(B)** $\frac{a^x}{\log a} + \frac{x^{a+1}}{a+1} + a^a x + C$

Watch out: Treating the constant a^a like x^a and writing $\frac{a^{a+1}}{a+1}$, option (C).

5 MCQ Board-pattern 1 mark

If $\int_0^{\pi/4} \tan^4 x\,dx = \alpha$, then $\int_0^{\pi/4} \tan^6 x\,dx$ is equal to:

(A) $\alpha - \frac{1}{5}$ **(B)** $\frac{1}{5} - \alpha$ **(C)** $\frac{1}{5} + \alpha$ **(D)** $\frac{1}{6} - \alpha$

Formula / idea: $\tan^n x + \tan^{n-2} x = \tan^{n-2} x \sec^2 x$; then put $\tan x = t$.

1 Add the two integrals and use $1 + \tan^2 x = \sec^2 x$

$$\int_0^{\pi/4} \tan^6 x\,dx + \alpha = \int_0^{\pi/4} \tan^4 x \sec^2 x\,dx$$

2 Put $\tan x = t$; limits 0 to 1

$$\int_0^1 t^4\,dt = \frac{1}{5}$$

3 Hence ✓ 1

$$\int_0^{\pi/4} \tan^6 x\,dx = \frac{1}{5} - \alpha$$

Answer **(B)** $\frac{1}{5} - \alpha$

Watch out: Writing $\frac{1}{6}$ from the power 6; after $\tan x = t$ the power left is t^4.

6 Assertion-Reason Board-pattern 1 mark

Assertion (A): $\int_{-\pi/2}^{\pi/2} \left(x^3 + x\cos x + \sin^5 x + 1\right) dx = \pi$. **Reason (R):** For every function f continuous on $[-a, a]$, $\int_{-a}^{a} f(x)\,dx = 2\int_0^a f(x)\,dx$.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).

(C) **Assertion (A) is true, but Reason (R) is false.**

(D) Assertion (A) is false, but Reason (R) is true.

Formula / idea: On $[-a, a]$: an odd function integrates to 0; only an even function gives $2\int_0^a f(x)\,dx$.

1 Check A: x^3, $x\cos x$, $\sin^5 x$ are odd, so their integrals are 0

$$\int_{-\pi/2}^{\pi/2} \left(x^3 + x\cos x + \sin^5 x + 1\right) dx = 0 + \int_{-\pi/2}^{\pi/2} 1\,dx = \pi$$

2 So A is true.

3	Check R with $f(x) = x$ on $[-1, 1]$ $\int_{-1}^{1} x\,dx = 0, \qquad 2\int_{0}^{1} x\,dx = 1$	
4	The two sides differ, so R is false (it holds only for even f).	
5	A false R cannot explain A.	
6	Hence the correct option is (C).	✓ 1

Answer **(C) A is true, but R is false**

Watch out: Marking (A) because R looks like a textbook property; it misses the condition that f must be even.

7 **Short answer** NCERT Misc. Ex. Q7 2 marks

Find: $\int \dfrac{\sin x}{\sin(x-a)}\,dx.$

Formula / idea: Write $x = (x-a) + a$ and expand $\sin(A+B)$.

1	Write $x = (x-a) + a$ $\sin x = \sin(x-a)\cos a + \cos(x-a)\sin a$	✓ ½
2	Divide by $\sin(x-a)$ $\dfrac{\sin x}{\sin(x-a)} = \cos a + \sin a \cot(x-a)$	✓ ½
3	Integrate; $\cos a$ and $\sin a$ are constants $I = \int \cos a\,dx + \sin a \int \cot(x-a)\,dx$	✓ ½
4	Use $\int \cot u\,du = \log\lvert\sin u\rvert$ $I = x\cos a + \sin a\,\log\lvert\sin(x-a)\rvert + C$	✓ ½

Answer $x\cos a + \sin a\,\log\lvert\sin(x-a)\rvert + C$

Watch out: Integrating $\cos a$ as $\sin a$; a is a constant, so $\int \cos a\,dx = x\cos a$.

Level 2

19 marks

8 **Short answer** Board-pattern 2 marks

Evaluate: $\int_0^{\pi/2} \cos x \cos 2x \cos 3x \, dx$.

Formula / idea: $2\cos A\cos B = \cos(A+B)+\cos(A-B)$, used twice.

1 Pair $\cos 3x$ with $\cos x$

$\cos x \cos 3x = \frac{1}{2}(\cos 4x + \cos 2x)$ ✓ ½

2 Multiply by $\cos 2x$ and use the identity again

$\cos x \cos 2x \cos 3x = \frac{1}{4}(\cos 6x + \cos 2x) + \frac{1}{4}(1 + \cos 4x)$ ✓ ½

3 Integrate

$I = \frac{1}{4}\left[\frac{\sin 6x}{6} + \frac{\sin 2x}{2} + x + \frac{\sin 4x}{4}\right]_0^{\pi/2}$ ✓ ½

4 Every sine term is 0 at $x = 0$ and at $x = \frac{\pi}{2}$

$\sin 3\pi = \sin \pi = \sin 2\pi = 0$

5 Hence

$I = \frac{1}{4} \cdot \frac{\pi}{2} = \frac{\pi}{8}$ ✓ ½

Answer $\frac{\pi}{8}$

Watch out: Writing $\cos 2x \cos 2x = \cos 4x$; it is $\cos^2 2x = \frac{1+\cos 4x}{2}$.

9 **Short answer** Board-pattern 2 marks

Given $\frac{d}{dx}F(x) = \frac{x}{\sqrt{1+2x^2-x^4}}$ and $F(1) = 0$, find $F(x)$.

Formula / idea: $\int \frac{dt}{\sqrt{a^2-t^2}} = \sin^{-1}\frac{t}{a} + C$

1 Complete the square in x^2

$1 + 2x^2 - x^4 = 2 - (x^2-1)^2$ ✓ ½

2 Put $x^2 - 1 = t$

$2x\,dx = dt \Rightarrow x\,dx = \frac{dt}{2}$ ✓ ½

3 Substitute and integrate with $a = \sqrt{2}$ ✓ ½

$$F(x) = \frac{1}{2}\int \frac{dt}{\sqrt{(\sqrt{2})^2 - t^2}} = \frac{1}{2}\sin^{-1}\left(\frac{x^2-1}{\sqrt{2}}\right) + C$$

4 Use $F(1) = 0$ ✓ ½

$$\frac{1}{2}\sin^{-1} 0 + C = 0 \;\Rightarrow\; C = 0, \qquad F(x) = \frac{1}{2}\sin^{-1}\left(\frac{x^2-1}{\sqrt{2}}\right)$$

Answer $F(x) = \frac{1}{2}\sin^{-1}\left(\frac{x^2-1}{\sqrt{2}}\right)$

Watch out: Completing the square as $2 + (x^2-1)^2$; take out the minus sign: $-(x^4 - 2x^2 - 1)$.

10 **Short answer** NCERT Ex 7.5 Q12 3 marks

Find: $\int \frac{x^3 + x + 1}{x^2 - 1}\,dx$.

Formula / idea: Degree of numerator $\geq$ degree of denominator: divide first, then use partial fractions.

1 Divide $x^3 + x + 1$ by $x^2 - 1$ ✓ ½

$$x^3 + x + 1 = x\left(x^2 - 1\right) + (2x + 1)$$

2 So ✓ ½

$$\frac{x^3 + x + 1}{x^2 - 1} = x + \frac{2x + 1}{(x-1)(x+1)}$$

3 Write as partial fractions ✓ ½

$$\frac{2x + 1}{(x-1)(x+1)} = \frac{A}{x-1} + \frac{B}{x+1} \;\Rightarrow\; 2x + 1 = A(x+1) + B(x-1)$$

4 Put $x = 1$, then $x = -1$ ✓ ½

$$3 = 2A \Rightarrow A = \frac{3}{2}, \qquad -1 = -2B \Rightarrow B = \frac{1}{2}$$

5 Integrate each term ✓ ½

$$I = \int x\,dx + \frac{3}{2}\int \frac{dx}{x-1} + \frac{1}{2}\int \frac{dx}{x+1}$$

6 Hence ✓ ½

$$I = \frac{x^2}{2} + \frac{3}{2}\log|x-1| + \frac{1}{2}\log|x+1| + C$$

Answer $\frac{x^2}{2} + \frac{3}{2}\log|x-1| + \frac{1}{2}\log|x+1| + C$

Watch out: Starting partial fractions without dividing; that misses the term x and gives wrong A, B.

11 **Short answer** Board-pattern 3 marks

Evaluate: $\int_{-\pi/2}^{\pi/2} \frac{\cos x}{1+e^x}\,dx.$

Formula / idea: $\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$; here $a+b=0$, so replace x by $-x$.

1 Let

$$I = \int_{-\pi/2}^{\pi/2} \frac{\cos x}{1+e^x}\,dx \quad \dots(1)$$

2 Replace x by $-\frac{\pi}{2}+\frac{\pi}{2}-x=-x$; $\cos(-x)=\cos x$ ✓ 1

$$I = \int_{-\pi/2}^{\pi/2} \frac{\cos x}{1+e^{-x}}\,dx = \int_{-\pi/2}^{\pi/2} \frac{e^x\cos x}{e^x+1}\,dx \quad \dots(2)$$

3 Add (1) and (2) ✓ ½

$$2I = \int_{-\pi/2}^{\pi/2} \frac{(1+e^x)\cos x}{1+e^x}\,dx = \int_{-\pi/2}^{\pi/2} \cos x\,dx$$

4 Integrate ✓ 1

$$2I = \left[\sin x\right]_{-\pi/2}^{\pi/2} = 1-(-1) = 2$$

5 Hence ✓ ½

$$I = 1$$

Answer 1

Watch out: Calling the integrand even or odd; it is neither, so you must add $f(x)$ and $f(-x)$.

12 **Short answer** NCERT Misc. Ex. Q24 3 marks

Evaluate: $\int_{\pi/2}^{\pi} e^x\left(\frac{1-\sin x}{1-\cos x}\right)dx.$

Formula / idea: $\int e^x\left[f(x)+f'(x)\right]dx = e^x f(x)+C$

1 Use $1-\cos x = 2\sin^2\frac{x}{2}$ and $\sin x = 2\sin\frac{x}{2}\cos\frac{x}{2}$ ✓ ½

$$\frac{1-\sin x}{1-\cos x} = \frac{1}{2\sin^2\frac{x}{2}} - \frac{2\sin\frac{x}{2}\cos\frac{x}{2}}{2\sin^2\frac{x}{2}}$$

2 Simplify ✓ ½

$$\frac{1-\sin x}{1-\cos x} = \frac{1}{2}\csc^2\frac{x}{2} - \cot\frac{x}{2}$$

3 Take $f(x) = -\cot\dfrac{x}{2}$
$f'(x) = \dfrac{1}{2}\csc^2\dfrac{x}{2}$ ✓ ½

4 Use the $e^x[f + f']$ result
$I = \left[-e^x \cot\dfrac{x}{2}\right]_{\pi/2}^{\pi}$ ✓ ½

5 Put the limits; $\cot\dfrac{\pi}{2} = 0$, $\cot\dfrac{\pi}{4} = 1$
$I = -e^{\pi}\cdot 0 + e^{\pi/2}\cdot 1 = e^{\pi/2}$ ✓ 1

Answer $e^{\pi/2}$

Watch out: Taking $f(x) = \cot\dfrac{x}{2}$; its derivative is $-\dfrac{1}{2}\csc^2\dfrac{x}{2}$, the wrong sign.

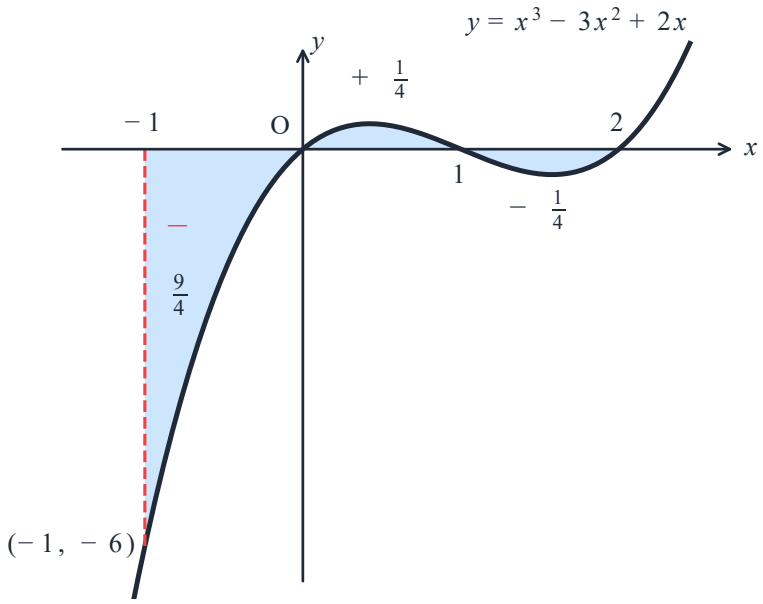
13 Short answer CBSE SQP 2021-22 (Term 2) 3 marks

Evaluate: $\displaystyle\int_{-1}^{2} \left|x^3 - 3x^2 + 2x\right| dx.$

Formula / idea: Split a modulus integral at the zeros of the expression inside and fix the sign on each piece.

1 Factorise
$x^3 - 3x^2 + 2x = x(x-1)(x-2)$ ✓ ½

2 Find the sign on each piece
$x(x-1)(x-2) \le 0$ on $[-1, 0]$, $\quad \ge 0$ on $[0, 1]$, $\quad \le 0$ on $[1, 2]$ ✓ ½



3 Split the integral; let $f(x) = x^3 - 3x^2 + 2x$
$I = -\displaystyle\int_{-1}^{0} f(x)\,dx + \int_{0}^{1} f(x)\,dx - \int_{1}^{2} f(x)\,dx$ ✓ ½

4 An antiderivative of f and its values

$$F(x) = \frac{x^4}{4} - x^3 + x^2; \qquad F(-1) = \frac{9}{4},\ F(0) = 0,\ F(1) = \frac{1}{4},\ F(2) = 0$$

✓ ½

5 Put the values in

$$I = -\left(0 - \frac{9}{4}\right) + \left(\frac{1}{4} - 0\right) - \left(0 - \frac{1}{4}\right) = \frac{9}{4} + \frac{1}{4} + \frac{1}{4}$$

✓ ½

6 Hence

$$I = \frac{11}{4}$$

✓ ½

Answer $\frac{11}{4}$

Watch out: Integrating $f(x)$ straight from -1 to 2 gives $-\frac{9}{4}$; the negative parts must be flipped.

14 Short answer NCERT Exemplar — 3 marks

Find: $\displaystyle\int \frac{\sin^6 x + \cos^6 x}{\sin^2 x \cos^2 x}\, dx.$

Formula / idea: $a^3 + b^3 = (a+b)^3 - 3ab(a+b)$ with $a = \sin^2 x$, $b = \cos^2 x$.

1 Use $a^3 + b^3 = (a+b)^3 - 3ab(a+b)$ and $\sin^2 x + \cos^2 x = 1$

$$\sin^6 x + \cos^6 x = 1 - 3\sin^2 x \cos^2 x$$

✓ 1

2 Divide by $\sin^2 x \cos^2 x$

$$\frac{\sin^6 x + \cos^6 x}{\sin^2 x \cos^2 x} = \frac{1}{\sin^2 x \cos^2 x} - 3$$

✓ ½

3 Write $1 = \sin^2 x + \cos^2 x$ in the first fraction

$$\frac{\sin^2 x + \cos^2 x}{\sin^2 x \cos^2 x} = \sec^2 x + \csc^2 x$$

✓ ½

4 Integrate

$$I = \int \left(\sec^2 x + \csc^2 x - 3\right) dx = \tan x - \cot x - 3x + C$$

✓ 1

Answer $\tan x - \cot x - 3x + C$

Watch out: Writing $\int \csc^2 x\, dx = \cot x$; it is $-\cot x$.

Level 3

27 marks

15 **Long answer** NCERT Example 42 5 marks

Evaluate: $\int_0^{\pi} \frac{x}{a^2\cos^2 x + b^2\sin^2 x}\,dx$, where $a > 0, b > 0$.

Formula / idea: $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$ removes x; $\int_0^{2a} f = 2\int_0^a f$ if $f(2a-x) = f(x)$.

1 Let

$$I = \int_0^{\pi} \frac{x}{a^2\cos^2 x + b^2\sin^2 x}\,dx \quad \dots(1)$$

2 Replace x by $\pi - x$; $\cos^2$ and $\sin^2$ do not change ✓ 1

$$I = \int_0^{\pi} \frac{\pi - x}{a^2\cos^2 x + b^2\sin^2 x}\,dx \quad \dots(2)$$

3 Add (1) and (2) ✓ ½

$$2I = \pi\int_0^{\pi} \frac{dx}{a^2\cos^2 x + b^2\sin^2 x}$$

4 The integrand is unchanged by $x \to \pi - x$, so use $\int_0^{2a} f = 2\int_0^a f$ ✓ ½

$$I = \pi\int_0^{\pi/2} \frac{dx}{a^2\cos^2 x + b^2\sin^2 x}$$

5 Split at $\frac{\pi}{4}$; divide by $\cos^2 x$ in the first, $\sin^2 x$ in the second ✓ ½

$$I = \pi\left[\int_0^{\pi/4} \frac{\sec^2 x\,dx}{a^2 + b^2\tan^2 x} + \int_{\pi/4}^{\pi/2} \frac{\csc^2 x\,dx}{a^2\cot^2 x + b^2}\right]$$

6 Put $\tan x = t$ in the first and $\cot x = u$ in the second ✓ 1

$$I = \pi\left[\int_0^1 \frac{dt}{a^2 + b^2t^2} - \int_1^0 \frac{du}{a^2u^2 + b^2}\right]$$

$d(\cot x) = -\csc^2 x\,dx$

7 Integrate ✓ 1

$$I = \frac{\pi}{ab}\left[\tan^{-1}\frac{bt}{a}\right]_0^1 - \frac{\pi}{ab}\left[\tan^{-1}\frac{au}{b}\right]_1^0 = \frac{\pi}{ab}\left[\tan^{-1}\frac{b}{a} + \tan^{-1}\frac{a}{b}\right]$$

8 Use $\tan^{-1} m + \tan^{-1}\frac{1}{m} = \frac{\pi}{2}$ for $m > 0$ ✓ ½

$$I = \frac{\pi}{ab}\cdot\frac{\pi}{2} = \frac{\pi^2}{2ab}$$

Answer $\frac{\pi^2}{2ab}$

Watch out: Putting $\tan x = t$ on all of $[0, \pi]$; $\tan x$ breaks at $\frac{\pi}{2}$, so reduce to $[0, \frac{\pi}{2}]$ first.

16 **Long answer** NCERT Misc. Ex. Q6 5 marks

Find: $\int \frac{5x}{(x+1)(x^2+9)}\,dx.$

Formula / idea: A factor x^2+9 that does not factorise needs a numerator $Bx+C$.

1 Write as partial fractions ✓ 1
$$\frac{5x}{(x+1)(x^2+9)} = \frac{A}{x+1} + \frac{Bx+C}{x^2+9}$$

2 Multiply by $(x+1)(x^2+9)$ ✓ ½
$$5x = A\left(x^2+9\right) + (Bx+C)(x+1)$$

3 Put $x=-1$ ✓ ½
$$-5 = 10A \;\Rightarrow\; A = -\frac{1}{2}$$

4 Compare coefficients of x^2 and constant terms ✓ 1
$$A+B=0 \Rightarrow B=\frac{1}{2}, \qquad 9A+C=0 \Rightarrow C=\frac{9}{2}$$

5 Split the integral ✓ ½
$$I = -\frac{1}{2}\int \frac{dx}{x+1} + \frac{1}{4}\int \frac{2x}{x^2+9}\,dx + \frac{9}{2}\int \frac{dx}{x^2+3^2}$$

6 Integrate each part ✓ 1
$$-\frac{1}{2}\log|x+1|, \qquad \frac{1}{4}\log\left(x^2+9\right), \qquad \frac{9}{2}\cdot\frac{1}{3}\tan^{-1}\frac{x}{3}$$

7 Hence ✓ ½
$$I = -\frac{1}{2}\log|x+1| + \frac{1}{4}\log\left(x^2+9\right) + \frac{3}{2}\tan^{-1}\frac{x}{3} + C$$

Answer $-\frac{1}{2}\log|x+1| + \frac{1}{4}\log\left(x^2+9\right) + \frac{3}{2}\tan^{-1}\frac{x}{3} + C$

Watch out: Taking only a constant $\frac{C}{x^2+9}$ over the quadratic factor; then no values of A, C fit.

17 **Long answer** NCERT Ex 7.10 Q16 5 marks

Evaluate: $\int_0^{\pi} \log(1+\cos x)\,dx.$

Formula / idea: $\int_0^{\pi/2} \log\sin x\,dx = \int_0^{\pi/2} \log\cos x\,dx = -\frac{\pi}{2}\log 2$

1 Use $1+\cos x = 2\cos^2\frac{x}{2}$ ✓ ½
$$I = \int_0^{\pi}\left[\log 2 + 2\log\cos\frac{x}{2}\right]dx = \pi\log 2 + 2\int_0^{\pi}\log\cos\frac{x}{2}\,dx$$

2 Put $\frac{x}{2} = t$, $dx = 2\,dt$; limits 0 to $\frac{\pi}{2}$ ✓ 1

$$I = \pi \log 2 + 4J, \qquad J = \int_0^{\pi/2} \log \cos t \, dt$$

3 Replace t by $\frac{\pi}{2} - t$ in J ✓ ½

$$J = \int_0^{\pi/2} \log \sin t \, dt$$

4 Add the two forms of J; use $\sin t \cos t = \frac{\sin 2t}{2}$ ✓ ½

$$2J = \int_0^{\pi/2} \log \frac{\sin 2t}{2} \, dt = \int_0^{\pi/2} \log \sin 2t \, dt - \frac{\pi}{2} \log 2$$

5 Put $2t = u$; limits 0 to π ✓ ½

$$\int_0^{\pi/2} \log \sin 2t \, dt = \frac{1}{2} \int_0^{\pi} \log \sin u \, du$$

6 Use $\int_0^{2a} f = 2 \int_0^{a} f$, as $\sin(\pi - u) = \sin u$ ✓ ½

$$\frac{1}{2} \int_0^{\pi} \log \sin u \, du = \int_0^{\pi/2} \log \sin u \, du = J$$

7 So ✓ ½

$$2J = J - \frac{\pi}{2} \log 2 \;\Rightarrow\; J = -\frac{\pi}{2} \log 2$$

8 Put J back ✓ ½

$$I = \pi \log 2 + 4\left(-\frac{\pi}{2} \log 2\right) = \pi \log 2 - 2\pi \log 2$$

9 Hence ✓ ½

$$I = -\pi \log 2$$

Answer $-\pi \log 2$

Watch out: Forgetting $dx = 2\,dt$ after $\frac{x}{2} = t$, which gives $2J$ instead of $4J$.

18 Short answer NCERT Exemplar 3 marks

Find: $\int \sqrt{\frac{a+x}{a-x}}\, dx$, where $a > 0$ and $-a < x < a$.

Formula / idea: Rationalise: multiply top and bottom by $\sqrt{a+x}$ to get $\frac{a+x}{\sqrt{a^2 - x^2}}$.

1 Multiply numerator and denominator by $\sqrt{a+x}$ ✓ ½

$$\sqrt{\frac{a+x}{a-x}} = \frac{a+x}{\sqrt{(a-x)(a+x)}} = \frac{a+x}{\sqrt{a^2 - x^2}}$$

2 Split

$$I = a\int \frac{dx}{\sqrt{a^2-x^2}} + \int \frac{x\,dx}{\sqrt{a^2-x^2}}$$

✓ ½

3 First integral: standard form

$$a\int \frac{dx}{\sqrt{a^2-x^2}} = a\sin^{-1}\frac{x}{a}$$

✓ ½

4 Second integral: put $a^2-x^2=t$, so $x\,dx=-\dfrac{dt}{2}$

$$\int \frac{x\,dx}{\sqrt{a^2-x^2}} = -\frac{1}{2}\int t^{-1/2}\,dt = -\sqrt{t} = -\sqrt{a^2-x^2}$$

✓ 1

5 Hence

$$I = a\sin^{-1}\frac{x}{a} - \sqrt{a^2-x^2} + C$$

✓ ½

Answer $a\sin^{-1}\dfrac{x}{a} - \sqrt{a^2-x^2} + C$

Watch out: Multiplying by $\sqrt{a-x}$ instead; that puts $a-x$ under a root in the numerator and gets nowhere.

19 **Long answer** NCERT Exemplar 5 marks

Find: $\displaystyle\int \sqrt{\tan x}\,dx.$

Formula / idea: $2t^2=(t^2+1)+(t^2-1)$; divide by t^2 and put $t-\dfrac{1}{t}=u$, $t+\dfrac{1}{t}=v$.

1 Put $\tan x = t^2$

$$\sec^2 x\,dx = 2t\,dt \;\Rightarrow\; dx = \frac{2t\,dt}{1+t^4}$$

✓ 1

2 Substitute

$$I = \int t\cdot\frac{2t}{1+t^4}\,dt = \int \frac{2t^2}{t^4+1}\,dt$$

✓ ½

3 Write $2t^2=(t^2+1)+(t^2-1)$

$$I = \int \frac{t^2+1}{t^4+1}\,dt + \int \frac{t^2-1}{t^4+1}\,dt = I_1 + I_2$$

✓ ½

4 In I_1 divide by t^2 and put $t-\dfrac{1}{t}=u$

$$I_1 = \int \frac{1+\frac{1}{t^2}}{\left(t-\frac{1}{t}\right)^2+2}\,dt = \int \frac{du}{u^2+2} = \frac{1}{\sqrt{2}}\tan^{-1}\frac{u}{\sqrt{2}}$$

✓ 1

5 In I_2 divide by t^2 and put $t+\dfrac{1}{t}=v$

$$I_2 = \int \frac{1-\frac{1}{t^2}}{\left(t+\frac{1}{t}\right)^2-2}\,dt = \int \frac{dv}{v^2-2} = \frac{1}{2\sqrt{2}}\log\left|\frac{v-\sqrt{2}}{v+\sqrt{2}}\right|$$

✓ 1

6 Put back $u = \dfrac{t^2-1}{t}$, $v = \dfrac{t^2+1}{t}$

$$I = \frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{t^2-1}{\sqrt{2}\,t}\right) + \frac{1}{2\sqrt{2}} \log\left|\frac{t^2-\sqrt{2}\,t+1}{t^2+\sqrt{2}\,t+1}\right| + C$$

✓ ½

7 Put back $t = \sqrt{\tan x}$

$$I = \frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{\tan x - 1}{\sqrt{2\tan x}}\right) + \frac{1}{2\sqrt{2}} \log\left|\frac{\tan x - \sqrt{2\tan x}+1}{\tan x + \sqrt{2\tan x}+1}\right| + C$$

✓ ½

Answer $\dfrac{1}{\sqrt{2}} \tan^{-1}\left(\dfrac{\tan x - 1}{\sqrt{2\tan x}}\right) + \dfrac{1}{2\sqrt{2}} \log\left|\dfrac{\tan x - \sqrt{2\tan x}+1}{\tan x + \sqrt{2\tan x}+1}\right| + C$

Watch out: Writing $dx = \dfrac{2t\,dt}{1+t^2}$; since $\sec^2 x = 1 + \tan^2 x = 1 + t^4$, it is $1 + t^4$.

20 Case study Board-pattern — 4 marks

Case Study: In a mathematics quiz, a team is told to use only the property $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$ (with standard integrals) to evaluate the integrals below. Answer the following.

(i) Evaluate $\displaystyle\int_0^{\pi/2} \log\left(\frac{4+3\sin x}{4+3\cos x}\right) dx.$ — 1 mark

(ii) Evaluate $\displaystyle\int_0^{2} x\sqrt{2-x}\,dx.$ — 1 mark

(iii) Evaluate $\displaystyle\int_0^{\pi/2} \frac{\sin^2 x}{\sin x + \cos x}\,dx.$ — 2 marks

Formula / idea: Replace x by $a - x$; then add the two forms of I, or integrate the simpler form.

(i) — 1 mark

1 Let

$$I = \int_0^{\pi/2} \log\left(\frac{4+3\sin x}{4+3\cos x}\right) dx$$

2 Replace x by $\dfrac{\pi}{2} - x$; sin and cos swap

$$I = \int_0^{\pi/2} \log\left(\frac{4+3\cos x}{4+3\sin x}\right) dx = -I$$

✓ ½

3 So

$$2I = 0 \;\Rightarrow\; I = 0$$

✓ ½

Answer (i) 0

(ii) 1 mark

1 Let

$$I = \int_0^2 x\sqrt{2-x}\,dx$$

2 Replace x by $2-x$; the root becomes $\sqrt{x}$ ✓ ½

$$I = \int_0^2 (2-x)\sqrt{x}\,dx = \int_0^2 \left(2x^{1/2} - x^{3/2}\right) dx$$

3 Integrate

$$I = \left[\frac{4}{3}x^{3/2} - \frac{2}{5}x^{5/2}\right]_0^2$$

4 Put the limits; $2^{3/2} = 2\sqrt{2}$, $2^{5/2} = 4\sqrt{2}$ ✓ ½

$$I = \frac{8\sqrt{2}}{3} - \frac{8\sqrt{2}}{5} = \frac{16\sqrt{2}}{15}$$

Answer (ii) $\dfrac{16\sqrt{2}}{15}$

(iii) 2 marks

1 Let

$$I = \int_0^{\pi/2} \frac{\sin^2 x}{\sin x + \cos x}\,dx \quad \ldots(1)$$

2 Replace x by $\dfrac{\pi}{2} - x$ ✓ ½

$$I = \int_0^{\pi/2} \frac{\cos^2 x}{\cos x + \sin x}\,dx \quad \ldots(2)$$

3 Add (1) and (2); $\sin^2 x + \cos^2 x = 1$ ✓ ½

$$2I = \int_0^{\pi/2} \frac{dx}{\sin x + \cos x}$$

4 Use $\sin x + \cos x = \sqrt{2}\sin\left(x + \dfrac{\pi}{4}\right)$ and $\displaystyle\int \csc\theta\,d\theta = \log|\csc\theta - \cot\theta|$ ✓ ½

$$2I = \frac{1}{\sqrt{2}}\left[\log\left|\csc\left(x+\frac{\pi}{4}\right) - \cot\left(x+\frac{\pi}{4}\right)\right|\right]_0^{\pi/2}$$

5 Put the limits; $\csc\dfrac{3\pi}{4} = \sqrt{2}$, $\cot\dfrac{3\pi}{4} = -1$

$$2I = \frac{1}{\sqrt{2}}\left[\log\left(\sqrt{2}+1\right) - \log\left(\sqrt{2}-1\right)\right] = \frac{1}{\sqrt{2}}\log\left(\sqrt{2}+1\right)^2$$

6 Hence ✓ ½

$$I = \frac{1}{2}\cdot\frac{2}{\sqrt{2}}\log\left(\sqrt{2}+1\right) = \frac{1}{\sqrt{2}}\log\left(\sqrt{2}+1\right)$$

$$\frac{\sqrt{2}+1}{\sqrt{2}-1} = \left(\sqrt{2}+1\right)^2$$

Answer (iii) $\dfrac{1}{\sqrt{2}} \log\left(\sqrt{2}+1\right)$

Watch out: In (iii), trying to integrate $\dfrac{\sin^2 x}{\sin x + \cos x}$ directly; add the $\cos^2 x$ form first.